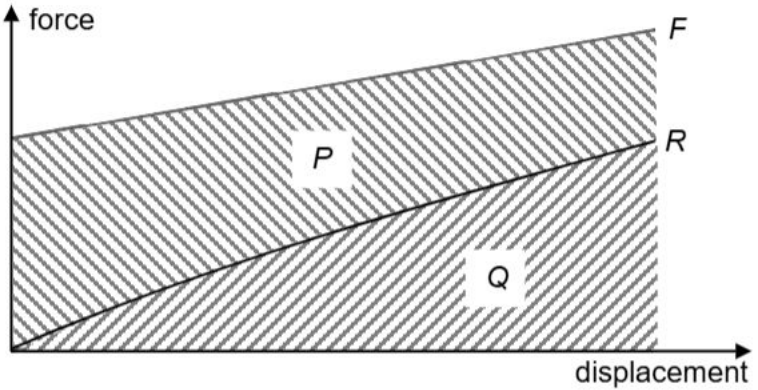
7 A body is accelerated from rest by a horizontal force F on a smooth surface through air. The magnitudes of F and air resistance R vary with displacement as shown in the graph below.



Which area represents the thermal energy dissipated?

A P **B** Q **C** $P - Q$ **D** $P + Q$

Ans: **B**

Thermal energy dissipated = Work done against air resistance R

= Area under force-displacement graph for R

8 An object is projected horizontally. The object falls a vertical distance y and travels a horizontal